MedAgent Clinical Decision Support

Case ID: case_60f2a51956 | **Status:** APPROVE | **Physician:** Dr. Alex Vance | **Date:** 2026-10-16 07:04 UTC

CRITICAL DISCLAIMER: Decision-support draft generated by AI agents. It is not a diagnosis and must be reviewed, corrected and approved by a licensed physician before any clinical use.

Patient Intake & Presentation

Patient: Age: Unspecified, Sex: Unspecified
Notes & History: .

Clinical Summary / Impression

Clinical evaluation indicates features consistent with Acute Hepatocellular Injury, characterized pathophysiologically by direct toxic, ischemic, or immunologic destruction of hepatocytes causing leakage of intracellular aminotransferases (alt, ast) into systemic circulation and acute hepatic metabolic decompensation. Key supporting clinical indicators include Hepatocellular Pattern, Hyperbilirubinaemia, while secondary differential considerations include Cholestasis / Obstructive Biliary Disease, Pleural Effusion. Targeted diagnostic workup and guided serial clinical monitoring are recommended to establish definitive management.

Ranked Differential Diagnoses

#	Condition	Likelihood	Key Supporting Findings	Citations
1	**Acute Hepatocellular Injury**	**MODERATE**	Hepatocellular Pattern, Hyperbilirubinaemia	S6, S7
2	**Cholestasis / Obstructive Biliary Disease**	**MODERATE**	Cholestatic Pattern, Hyperbilirubinaemia	S4, S8
3	**Pleural Effusion**	**MODERATE**	Effusion (p=68.1%), Hypoalbuminaemia	S1, S5

Radiographic Findings (Chest X-Ray)

Pathology / Finding	Probability	Location	Indicative Severity	Confidence
Effusion	**68.1%**	right lower zone	Moderate	Moderate

Laboratory Findings & Identified Patterns

Analyte / Test	Result	Unit	Reference Interval	Flag
ALT (SGPT)	**580**	U/L	7 – 56	**CRITICAL HIGH**
AST (SGOT)	**620**	U/L	10 – 40	**CRITICAL HIGH**
Total Bilirubin	**5.4**	mg/dL	0.2 – 1.2	**HIGH**
Direct Bilirubin	**3.8**	mg/dL	0 – 0.3	**HIGH**
Alkaline Phosphatase	**210**	U/L	44 – 147	**HIGH**
Serum Albumin	**3.1**	g/dL	3.5 – 5	**LOW**

Inferred Diagnoses (ClinicalLabNet Neural Classifier):
• **[SEVERE] Acute Hepatocellular Injury / Transaminitis** [100.0% Neural Confidence] (Likelihood: HIGH): Neural Network Confidence: 99.9% (ClinicalLabNet deep multi-label classifier). Biomarker pattern is characteristic of acute hepatocellular injury / transaminitis. *Pathophysiology:* Direct toxic, ischemic, or viral destruction of hepatocytes resulting in extensive leakage of cytosolic aminotransferases into circulation.
• **[MODERATE] Cholestasis / Obstructive Biliary Disease** [98.5% Neural Confidence] (Likelihood: HIGH): Neural Network Confidence: 98.5% (ClinicalLabNet deep multi-label classifier). Biomarker pattern is characteristic of cholestasis / obstructive biliary disease. *Pathophysiology:* Impairment of bile excretion secondary to extrahepatic ductal stone/stricture or intrahepatic canalicular injury, causing accumulation of bile acids, alkaline phosphatase, and conjugated bilirubin.

Synthesized Clinical Patterns:
• **[SOON] Hepatocellular Pattern:** Marked transaminase elevation reflecting acute hepatocellular injury or necrosis.
• **[SOON] Cholestatic Pattern:** Elevation of alkaline phosphatase, GGT, or direct bilirubin indicating biliary stasis or obstruction.
• **[SOON] Hyperbilirubinaemia:** Elevated total bilirubin with visible or subclinical jaundice.
• **[ROUTINE] Hypoalbuminaemia:** Low serum albumin reflecting systemic inflammation, malnutrition, nephrosis, or reduced hepatic synthesis.

Suggested Diagnostic Workup

• Viral hepatitis serologies (HAV, HBV, HCV)
• Serum acetaminophen level and toxicology screen
• Coagulation profile (PT/INR)

• Right upper quadrant abdominal ultrasound
• Diagnostic thoracentesis with Light's criteria

Physician Attestation: I, Dr. Alex Vance, hereby confirm that I have reviewed, verified, and adjusted this decision-support synthesis. Decision: **APPROVE**.
Reviewer Notes: No specific notes recorded.
Signature: _____ **Date:** 2026-10-16 07:04 UTC